Assignment no 10:

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle,
Perimeter = 2*(L+W) and Area = L*W.

```
d2_96695_sahil> create procedure sp_area_perim(In length double,In width double,OUT perimeter double,OUT area double)
    -> begin
    -> SET perimeter=2*(length+width);
    -> SET area=length*width;
    -> END
    -> $$
Query OK, 0 rows affected (0.01 sec)

d2_96695_sahil> call sp_area_perim(55,4,@p,@r) $$
Query OK, 0 rows affected (0.00 sec)

d2_96695_sahil> select @p as perimeter,@r as area$$
+-----------+------+
| perimeter | area |
+-----------+------+
|       118 |  220 |
+-----------+------+
1 row in set (0.00 sec)

d2_96695_sahil> |
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
d2_96695_sahil> create table temp (num int,num_square int,num_cube int);
Query OK, 0 rows affected (0.03 sec)

d2_96695_sahil> desc temp;
+------------+------+------+-----+---------+-------+
| Field      | Type | Null | Key | Default | Extra |
+------------+------+------+-----+---------+-------+
| num        | int  | YES  |     | NULL    |       |
| num_square | int  | YES  |     | NULL    |       |
| num_cube   | int  | YES  |     | NULL    |       |
+------------+------+------+-----+---------+-------+
3 rows in set (0.00 sec)

d2_96695_sahil> delimiter $$
d2_96695_sahil> CREATE PROCEDURE num_sc(numm INT)
    -> begin
    -> DECLARE num INT;
    -> DECLARE num_square INT;
    -> DECLARE num_cube INT;
    -> SELECT numm INTO num;
    -> SELECT numm*numm INTO num_square;
    -> SELECT numm*numm*numm INTO num_cube;
    -> Insert into temp values (num,num_square,num_cube);
    -> END;
    -> $$
Query OK, 0 rows affected (0.01 sec)

d2_96695_sahil> call num_sc(5)$$
Query OK, 1 row affected (0.01 sec)

d2_96695_sahil> select * from temp$$
+------+------------+----------+
| num  | num_square | num_cube |
+------+------------+----------+
|    5 |         25 |      125 |
+------+------------+----------+
1 row in set (0.00 sec)

d2_96695_sahil> |
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are
C= (F-32) *5/9 F= 9/5*C + 32

```
d2_96695_sahil> CREATE PROCEDURE sp_temp(IN p_temp DOUBLE,IN p_unit CHAR)
    -> BEGIN
    -> DECLARE v_result DOUBLE;
    -> CASE
    -> WHEN (p_unit = 'c' OR p_unit = 'C')
    -> then SET v_result = ((9/5*p_temp)+32);
    -> SELECT CONCAT(v_result, '°F') AS 'Temperature';
    -> WHEN (p_unit = 'f' OR p_unit = 'F')
    -> then set v_result = ((p_temp -32)*5/9);
    -> SELECT CONCAT(v_result, '°C') AS 'Temparature';
    -> END CASE;
    -> END;
    -> $$
Query OK, 0 rows affected (0.01 sec)

d2_96695_sahil> call sp_temp(100,'C');
    -> $$
+-------------+
| Temperature |
+-------------+
| 212°F       |
+-------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

d2_96695_sahil> call sp_temp(40,'F');
    -> $$
+---------------------+
| Temparature         |
+---------------------+
| 4.444444444444445°C |
+---------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

d2_96695_sahil> DELIMITER ;
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
d2_96695_sahil> USE CLASSWORK $$
Database changed
d2_96695_sahil> CREATE PROCEDURE convertInches(value INT)
    ->      BEGIN
    ->      DECLARE yards INT;
    ->      DECLARE foot INT;
    ->      DECLARE inch INT;
    ->      SET yards = value/36;
    ->      SET foot =(value%36)/12;
    ->      SET inch=(value %36)%12;
    ->      SELECT yards,foot,inch;
    ->      END;
    ->     $$
Query OK, 0 rows affected (0.08 sec)

d2_96695_sahil> DELIMITER ;
d2_96695_sahil> CALL convertINCHES(124);
+-------+------+------+
| yards | foot | inch |
+-------+------+------+
|     3 |    1 |    4 |
+-------+------+------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

d2_96695_sahil>
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
d2_96695_sahil> delimiter $$
d2_96695_sahil> CREATE PROCEDURE numm(num1 INT,num2 INT)
    -> begin
    -> declare n_result INT;
    -> select num1* num2 INTO n_result;
    -> if(n_result>=100) THEN
    -> SELECT 'product of two numbers is equal or greater than 100' as msg;
    -> else
    -> SELECT 'product of two numbers is not equal or greater than 100' as msg;
    -> end if;
    -> end;
    -> $$
Query OK, 0 rows affected (0.01 sec)

d2_96695_sahil> call numm(34,45)$$
+-----------------------------------------------------+
| msg                                                 |
+-----------------------------------------------------+
| product of two numbers is equal or greater than 100 |
+-----------------------------------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

d2_96695_sahil> call numm(4,5)$$
+-------------------------------------------------------+
| msg                                                   |
+-------------------------------------------------------+
| product of two numbers is not equal or greater than 100 |
+-------------------------------------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

d2_96695_sahil>
```